International Journal of Information Management

Contents lists available at ScienceDirect

International Journal of Information Management

journal homepage: www.elsevier.com/locate/ijinfomgt

Digital World: Influence of Biomedical Content and Device Type on Interaction with Pharmaceutical Ads

Isaac Mena [a,*]

[a] University of Science and Technology, *San José Province, San José · 4003 1391*

ARTICLE INFO

Keywords:
Data analysis
Biostatistics
Biomedical marketing
Digital interaction
Health advertising

ABSTRACT

The growing digitalization of the biomedical and pharmaceutical sectors has reshaped how users access health information, creating both opportunities and challenges for reliable content dissemination. This study addresses the problem of whether the amount of biomedical information in digital ads influences user interaction, a critical issue in an environment saturated with messages and prone to misinformation. The research is grounded in theories such as the Elaboration Likelihood Model (ELM) and heuristics in decision-making, which explain how message technicality and clarity impact consumer perception and behavior. A quantitative, non-experimental, and correlational design was applied, combining descriptive analysis, regression, and ANOVA. Data were collected from Google Campaign Manager 360 and DoubleVerify, analyzing active pharmaceutical campaigns in 2024. Results show that ads with lower biomedical content tended to generate more impressions and clicks, although these differences were not statistically significant. Similarly, the type of device had no significant effect on interaction rates. These findings suggest that message simplicity and clarity may be more influential than the amount of biomedical information provided. In conclusion, this study contributes evidence to optimize health communication campaigns, highlighting the importance of combining concise, appealing, and reliable data-driven messages to enhance both accessibility and effectiveness in health information dissemination.

1. Introduction

Reliable biomedical data are essential for the prevention, diagnosis, and treatment of diseases. Yet, as healthcare continues to digitalize, the way this information is delivered has become just as important as the data itself. For instance, Yltävä (2024) reported that in 2022 nearly a quarter of U.S. respondents purchased medications online at least once a month, compared to only 15% in Germany. These differences highlight not only the global relevance of digital access to health products, but also the urgent need for trustworthy, evidence-based information to guide such decisions.

Despite the growth of digital strategies to promote medical treatments, there are still considerable challenges in how biomedical content is disseminated. Research by Gamboa-Peñaranda et al. (2014) suggests that factors such as placement, design, and the sheer amount of information in digital biomedical advertisements remain poorly understood in terms of their influence on consumer behavior. We know that the biomedical and pharmaceutical sectors generate vast amounts of data about user interactions—impressions, clicks, engagement rates, and even behavioral changes. What we do not yet fully grasp is how these interactions are shaped by the informational depth of the content itself, and how that, in turn, affects health decision-making.

Marketing theory has long stressed the critical role of communication in shaping consumer perceptions. Kotler and Keller (2012), for example, emphasize that in markets where information is highly accessible but equally prone to misinformation, communication strategies must not only persuade but also build trust.

* Corresponding author.
E-mail address: imenam283@ulacit.ed.cr

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More recent work reinforces this view: Fierro Moreno and Hernández Carreón (2017) describe how pharmaceutical marketing is entangled with ethical responsibilities, while Smith et al. (2023) highlight how digitalization enables personalization of health content, offering opportunities to make medical information both more reliable and more accessible. However, the literature also points to persistent risks: without careful design, biomedical ads may either overwhelm audiences with technical detail or oversimplify in ways that foster misinformation (Fernández et al., 2021; Johnson & Brown, 2022).

This tension reveals a knowledge gap at the intersection of marketing and biomedical communication: we know that digital platforms can shape user engagement, but we lack empirical clarity on how the amount of biomedical information included in an ad influences user interaction. Martínez et al. (2023) argue that biomedical data analytics has become a promising field precisely because of these unresolved questions, and Pérez et al. (2022) note that statistical tools such as regression and correlation are uniquely suited to uncovering patterns of behavior in large datasets.

The present study seeks to contribute to this conversation by examining whether comprehensive biomedical content in digital advertisements enhances or hinders user engagement. By analyzing large-scale data from pharmaceutical campaigns in 2024 using Google Campaign Manager, this research offers insights that are not only theoretically valuable but also practically relevant. On a theoretical level, it extends our understanding of how marketing strategies intersect with public health communication. Methodologically, it demonstrates the application of biostatistical tools to real-world datasets. Practically, it speaks to the design of more effective, ethical, and evidence-based digital campaigns that can support both commercial objectives and public health goals (Gómez & Ramírez, 2021; Sánchez & López, 2023).

The central research question guiding this study is: How relevant is comprehensive biomedical content in digital advertisements to consumers' purchasing decisions in 2024? In addressing this question, the study aims to identify patterns of user interaction with digital content related to medical treatments, evaluate how content format relates to engagement, and assess whether device type influences click-through rates. Taking together, these objectives aim to bridge the gap between marketing practice and biomedical communication, advancing a more critical, data-driven understanding of how information shapes behavior in the digital health landscape.

2. Methodological Framework

2.1 Research Approach Type

2.1.1 Type of Research Approach

The approach of this research is quantitative, as it is based on the collection and analysis of numerical data with the aim of drawing conclusions grounded in empirical evidence. According to Creswell (2018), the quantitative approach is characterized by the objective measurement of variables and the use of statistical tools to analyze the relationships between them. In this study, statistical analysis methods will be applied to evaluate user interaction with biomedical advertisements on digital platforms, which will allow the identification of trends and patterns based on predefined variables.

Additionally, this approach follows a non-experimental design, as the data will be collected without actively manipulating the variables. According to Creswell (2018), this allows for observing effects in a natural manner. The research will also employ structured data collection techniques, such as the measurement of digital metrics (impressions, clicks, and conversions), aiming to produce results that are replicable and generalized.

2.1.2 Type of Scientific Research

The objective is to identify patterns of association between variables, which establishes the study as correlational, as it allows for the exploration of relationships between different factors without asserting causality. Correlational studies are fundamental for data analysis in statistical contexts, enabling the identification of connections between variables based on empirical evidence. In this case, the relationship between the amount of information in advertisements and user interaction on digital platforms will be analyzed (Umakantha, 2016).

Additionally, the research is also considered descriptive, as it focuses on observing and documenting the characteristics of phenomena without intervening in them, which makes it particularly useful for exploratory studies in the digital environment. As explained by D'Andrea and Wooten (2017), the use of descriptive analyses in statistical studies allows for the understanding of behavioral trends and patterns based on quantifiable data, providing relevant information on consumer responses to different formats of biomedical advertisements.

2.2 Type of Research Design

The research will adopt a cross-sectional and quantitative design to analyze the relationship between the amount of information in biomedical advertisements and user interaction on digital platforms. The cross-sectional design is characterized by the collection of data at a single point in time, allowing for a "snapshot" of the variables of interest without the need for prolonged follow-up. According to Babbie (2010), cross-sectional studies are useful for describing patterns and associations at a specific moment, facilitating the analysis of social and behavioral phenomena without requiring longitudinal observations.

Metrics such as impressions, clicks, and conversions will be analyzed over a specific period, providing information on how consumers interact with different formats of biomedical advertisements. Additionally, the quantitative nature of the study involves the application of statistical methods to examine the relationships between the previously mentioned variables (Babbie, 2010).

2.3 Practical Area of the Study

This study is classified as a field study, as the data will be obtained from real user interactions with digital advertisements on platforms such as Google Campaign Manager and DoubleVerify. This approach allows for the analysis of consumer behavior in a natural environment, without the influence of a laboratory or controlled conditions, which enhances the validity of the results (Google, 2024).

According to Guzmán (2019), field studies are essential for understanding phenomena in their real context, as they allow for data collection in non-artificial environments and the evaluation of variables under authentic conditions. In this case, metrics such as impressions, clicks, and conversion rates in biomedical advertisements will be observed, providing key information on how the amount of information influences user interaction.

2.4 Population and Sample

The population consists of the indirect participants in this study: internet users who interacted with advertisements on various devices (smartphones, desktop computers, tablets), although no personal information is collected and there is no direct contact with them. Therefore, the analysis is limited to the aggregate and anonymous behavior of users, adhering to ethical principles of confidentiality and data privacy.

2.4.1 Definition of the Population and the Sample

The sample used corresponds to a subset of data derived from active campaigns in 2024. These data include interaction metrics such as impressions, clicks, and conversions, segmented by device type (desktop computer, mobile phone, tablet). The analysis focuses on the statistical unit represented by each recorded interaction with the advertisements, allowing for the evaluation of variables such as device type, the amount of biomedical information included in the ads, and click behavior. This strategy aligns with the correlational and descriptive objectives of the study and enables the identification of significant associations based on quantifiable evidence without directly intervening in the target population (Sampieri, Collado & Lucio, 2014).

2.4.1.1 Type of Sampling

The sampling strategy used is non-probabilistic by convenience, as it was based on a pre-existing dataset from active campaigns during the selected period. This technique is appropriate for descriptive and correlational studies when the goal is to obtain information from accessible sources that reflect real behaviors in a digital context (Hernández Sampieri, Mendoza Torres & Baptista Lucio, 2021).

2.4.1.2 Sample Size Calculation

The sample size was determined through criterion-based non-probabilistic sampling, intentionally selecting the 100 advertisement lines that recorded the highest number of impressions during the active campaigns of 2024. This methodological decision corresponds to the descriptive and correlational nature of the study, which prioritizes the quality and relevance of the data over the randomness of selection. This strategy seeks to maximize the representativeness of the most viewed content, enabling a more robust analysis of user interaction according to device type and the amount of biomedical information present in the advertisements (Sampieri, Collado & Lucio, 2014).

2.4.1.3 Data Collection Strategy

Data collection in this study will be conducted using digital analytics tools such as Google Campaign Manager 360 and DoubleVerify, which allow for capturing key metrics such as impressions, clicks, and conversion rates.

elow is a detailed step-by-step description of the process followed by Campaign Manager 360 for data collection, in accordance with the measurement methodologies recommended by the Interactive Advertising Bureau (IAB):

Step-by-step data capture process:

1. The user accesses content (such as a mobile application or a website) that contains ad tags from Campaign Manager 360.
2. The application or browser sends a request to Campaign Manager 360 to retrieve an ad.
3. Campaign Manager 360 selects and delivers the creative content corresponding to the ad.
4. The creative content includes a code that, when the ad download begins, calls the Campaign Manager 360 ad server.
5. In the case of mobile applications enabled with OMID (Open Measurement SDK), an impression is counted when at least one pixel of the ad appears on the device screen.
6. Campaign Manager 360 registers the impression, although it may be filtered as invalid if an error or anomaly is detected in the process.
7. Finally, the ad content is delivered to the user by the Campaign Manager 360 server or a third-party server.

This procedure ensures that data capture aligns with international standards and provides reliable metrics that will allow for the analysis of user interaction with biomedical advertisements on digital platforms. The metrics mentioned are then exported directly from Campaign Manager reports (Google, 2024)

2.5 Practical Area of the Study

Objective	Variable or Category	Conceptual Definition	Operational Definition
Identify whether there are user interaction patterns with ads and digital materials about medical treatments.	**Impressions**	Number of times an ad is displayed on a digital platform.	Measured through reports generated by Google Campaign Manager 360 that register every time an ad loads on screen.
Identify whether there are user interaction patterns with ads and digital materials about medical treatments.	**Clicks**	Number of times users click on the ad.	Counted using click data from Google Campaign Manager 360 and DoubleVerify for each displayed ad.
Analyze the influence of the device type on the click-through rate of each digital ad.	**Inventory Type**	Classification of ads by digital channel (display, video, social media, etc.).	Categorized based on reports from the ad management system, identifying the media type used per ad line.
Analyze the influence of the device type on the click-through rate of each digital ad.	**Dimension**	Placement of the ad on the platform and segmentation features.	Extracted from digital media breakdowns, classifying locations (top, sidebar, mid-content) and applied segmentations (audience, behavior, device, etc.).
Evaluate the relationship between the biomedical content format and user engagement rate.	**Amount of Biomedical Information per Creative**	Degree of technicality and detail in biomedical concepts used in the ad.	Coded through content analysis as: low (common language), medium (some technical terms), or high (extensive biomedical language and scientific evidence).

Table 1. Practical Area of the Study (Author's own work, 2025.)

2.6 Research Techniques and Procedures

In studies involving the analysis of digital campaigns, the use of specialized measurement and verification tools is essential to ensure the validity and reliability of the data collected. In this study, platforms recognized in the digital advertising industry were used: Google Campaign Manager 360 and DoubleVerify. These platforms provide structured and auditable data without the need for traditional data collection techniques such as surveys or interviews (Garfinkel, 2021).

Google Campaign Manager 360 is a digital campaign management platform that offers advanced tools for ad delivery, tracking, and reporting. This tool not only allows for the distribution of advertising content but also enables the retrieval of key metrics such as impressions, clicks, conversion rates, device type, and ad placement. One of its main methodological contributions is the generation of standardized reports that comply with international standards set by the Interactive Advertising Bureau (IAB), ensuring that the data are comparable and technically reliable (Google, n.d.).

For its part, DoubleVerify acts as an independent verification system that validates the quality and legitimacy of advertising traffic. Through its technology, it is possible to verify ad visibility, detect advertising fraud (such as bots or non-human clicks), and ensure that the content is displayed in safe and appropriate environments. According to Garfinkel (2021), platforms like DoubleVerify fulfill an essential function in the digital value chain, as they increase ecosystem transparency and enable decision-making based on clean and reliable data.

2.6.1 Techniques or Instruments

The data will be obtained through analytical reports from the specialized platforms: Google Campaign Manager 360 and DoubleVerify. These platforms provide structured data on the performance of digital campaigns, including metrics such as impressions, clicks, conversions, inventory type, segmentation, device, and creative content. These metrics are included in Excel reports in CSV format, accompanied by valuable information such as the names of the "Placements," their unique "Placement ID," and the creative associated with each Placement. This makes it possible to analyze the performance of each digital ad across more than 1,000 different web pages simultaneously, since the impressions/clicks per ad are recorded in detail by the platforms (Miccoli, Compagnoni, Spinelli, & Lacaita, 2011).

Google Campaign Manager 360 is a management and measurement tool for advertising campaigns that provides accurate metrics, such as the number of impressions, clicks, and conversions per ad. The extracted reports allow for data segmentation by variables such as device type, ad placement, and creative content. DoubleVerify complements this analysis by verifying ad visibility, validating impressions, and detecting digital fraud, providing an additional layer of integrity and accuracy to the data collected (Ilyas, 2020).

2.6.2 Procedures

Data collection was carried out through the following steps. This procedure was performed in a digital environment, using data obtained from real user interactions on web platforms, without direct intervention on the sample. The analysis was limited to the year 2024.

1. **Selection of relevant campaigns**: Active biomedical campaigns in 2024 were identified within the managed advertising account. All campaigns in the account belong to a pharmaceutical client and include ads ranging from videos with little biomedical information to HTML5 ads with information on the patients involved in clinical trials.
2. **Report extraction**: Reports were generated from Google Campaign Manager 360 and DoubleVerify, prioritizing metrics such as impressions, clicks, device type, inventory type, dimensions, and creative content.

Campaign	Placement	Placement ID	Impressions	Clicks	Total Conversions
TEST CAMPAIGN A	0x0 video test	398311982	2	0	0
TEST CAMPAIGN A	160x600 test	391911753	12	0	0
TEST CAMPAIGN A	300 × 50 test	391639685	12	0	0
TEST CAMPAIGN A	300x250 test	391604896	9	0	0
TEST CAMPAIGN A	300x600 test	391641308	12	0	0
TEST CAMPAIGN A	320 × 50 test	391605457	10	0	0
TEST CAMPAIGN A	728 × 90 test	391604947	10	0	0
TEST CAMPAIGN A	video training 11	407981477	9	0	0
Total		Placement ID - N/A	76	0	0

Illustration 1 – Screenshot of the CM platform (Author's own work, 2025)

3. **Sampling criteria**: In both reports, creatives with the highest impressions/clicks were filtered from highest to lowest, to analyze why these were the 100 ad lines with the greatest number of impressions, following a non-probabilistic convenience sampling method.
4. **Biomedical content coding**: The 100 creatives were manually analyzed to classify the level of biomedical information based on a rubric of low, medium, or high complexity. This considered whether the content included ISI information about patients involved in clinical trials or merely the minimum required for FDA approval.
5. **Data organization**: All data were systematized in a spreadsheet, with variable identification and corresponding coding for statistical analysis, which are included in the annexes.

2.7 Data Analysis and Processing

According to the article by Google (2024), the Interactive Advertising Bureau (IAB) has published industry-wide standards for counting impressions and clicks. Campaign Manager 360 uses counting methodologies that comply with the latest version of IAB guidelines and undergoes regular third-party audits to ensure the accuracy and integrity of its counting and billing processes. Campaign Manager 360 counts an impression as part of the ad delivery process as follows:

1. A user accesses content that contains Campaign Manager 360 ad tags, such as a mobile application or a website.
2. The application or browser sends a request to Campaign Manager 360 to retrieve an ad.
3. Campaign Manager 360 selects an ad and delivers the creative content.
4. The creative content includes a code that calls the Campaign Manager 360 ad server once the content begins downloading.
5. For ad traffic in mobile apps enabled with OMID, Campaign Manager 360 and Display & Video 360 count an impression when at least one pixel of the ad appears on the device screen.
6. Campaign Manager 360 records the impression (which may later be filtered as invalid).
7. The creative content is delivered by Campaign Manager 360 or a third-party ad server.

Specifically for this research, the inventory type provided by DoubleVerify will be used as a complement to the Campaign Manager data, as this will help determine whether the device type or the platform where the ad is viewed influences conversion or impression rates. The type of device and characteristics of the digital inventory play a crucial role in determining conversion and impression rates, as they affect how users interact with ads (Chaffey, 2019).

Device Delivery Type	Authentic	Pinterest
	Attention	Reddit
	GroupM Billable	Snapchat
	Netflix	Standard
	Meta	TikTok
		Twitter
		YouTube

Table 2. Device Delivery Type (DoubleVerify, 2025.)

According to Wimmer & Dominick (2016), descriptive analysis is essential to obtain a clear view of the data and understand user interactions with the ads. By using measures of central tendency and dispersion, key patterns and behaviors can be identified that may be improved or replicated in future campaigns. This is the first step in understanding interaction patterns in the collected data. The following descriptive statistics will be used:

- **Mean**: To obtain the average number of impressions, clicks, or conversions.
- **Standard deviation**: To measure the variability of the data.
- **Frequency**: To count how often certain interactions or behaviors are repeated.
- **Percentages and proportions**: To compare the performance of different ads based on content type.

Correlation analysis—specifically **Pearson's correlation**—is, according to Field (2013), a valuable technique to assess whether a significant relationship exists between continuous variables, such as the amount of information in ads and interaction rates (clicks or conversions). This analysis allows for quantifying the linear relationship between two continuous variables. A value close to 1 or -1 indicates a strong relationship, while a value near 0 indicates little or no relationship.

With the help of the **Minitab** tool, advanced statistical techniques were applied to evaluate complex relationships between variables and optimize ad performance. Montgomery (2017) describes regression models as useful for predicting the impact of different factors (such as ad placement or audience segmentation) on conversion rates. He also mentions that statistical methods such as **analysis of variance (ANOVA)** allow for evaluating whether the differences in conversion rates between groups of ads are statistically significant.

2.8 Primary and Secondary Sources of Research

2.8.1 Primary Sources of Research

The primary sources in this context are the direct data obtained from digital platforms such as Google Campaign Manager and DoubleVerify, which allow for detailed information about user interactions with biomedical advertisements. These data include:

- **Impressions**: Number of times the ads are displayed to users.
- **Clicks**: Number of times users interact with the ads, either by clicking on them or showing interest in the content.
- **Conversion rates**: Proportion of users who perform the desired action (such as a purchase, registration, or inquiry about the ad content) after clicking.

A study by Kumar & Shah (2020) on digital advertising in the healthcare sector shows that the use of primary data from ad-serving platforms, such as Google Campaign Manager and DoubleVerify, is essential for optimizing advertising campaigns. Interaction data (impressions, clicks, conversion rates) form the basis for adjusting targeting, ad creativity, and distribution strategy. Analyzing these data makes it possible to identify user behavior patterns, leading to more effective resource allocation and continuous improvement in the dissemination of biomedical content in advertising campaigns.

2.8.2 Secondary Information Sources

1. **Digital Advertising Strategies in Health**
   A key study on digital advertising in the health sector is the work by Baker et al. (2017), which examines best practices for audience segmentation in digital marketing campaigns targeting healthcare professionals and consumers. This study provides insights into how segmentation strategies can affect conversion rates and the overall performance of advertising campaigns in the health sector.
2. **Health Advertising: Campaign Effectiveness**
   The article by Liu et al. (2018) analyzes the effectiveness of digital advertising campaigns in the health sector, focusing on the factors that determine conversion rates in ads related to medical products and services. It also addresses the importance of using accurate data to refine campaigns and improve conversion outcomes.
3. **Studies on Segmentation in Digital Advertising**
   Chaffey (2019) discusses how audience segmentation plays a crucial role in the success of digital advertising campaigns. The author suggests that understanding user preferences and behaviors—based on previous data—is key to personalizing ads and improving conversion rates.

3. Results

In order to understand the dynamics between the amount of biomedical content present in digital ads and user interaction with them, various statistical tools aimed at quantitative data analysis were applied. These included regression models, ANOVA tests, and descriptive analyses, with the goal of evaluating patterns in impressions and clicks based on the type of device used and the level of technicality in the creatives. The application of these techniques makes it possible to explore general trends, as well as determine the existence (or not) of significant differences among the different levels of medical information presented in the advertising pieces.

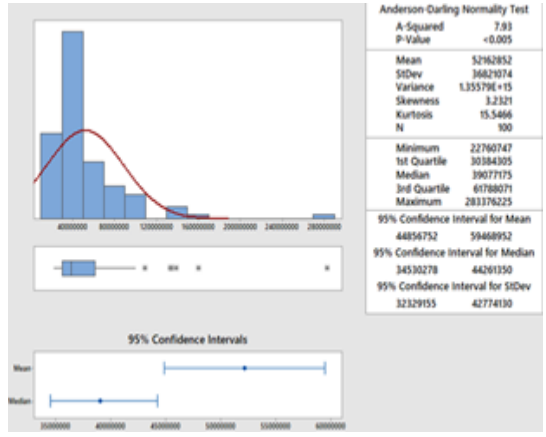

Illustration 6: Normal probability plot for the residuals of the regression model (impressions). (Author's elaboration, 2025)

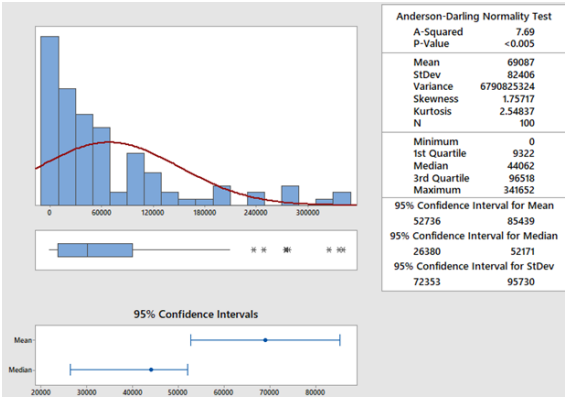

Illustration 7: Normal probability plot for the residuals of the regression model (clicks). (Author's elaboration, 2025)

Illustrations 6 and 7 correspond to an Anderson-Darling normality test applied to two different datasets, related through metrics such as clicks and impressions. In both cases, the p-value is less than 0.005, indicating a significant violation of normality in both distributions (Anderson & Darling, 1954).

In the left panel, the data show strong positive skewness (skewness = 1.75717) and elevated kurtosis (2.54837), suggesting a skewed distribution with outliers to the right. This is visually confirmed in the histogram, which shows a buildup of low values and a long right tail, as well as in the box plot where multiple upper outliers are identified. The 95% confidence intervals for the mean (52,736 to 85,439) and the median (26,380 to 52,171) do not completely overlap, reinforcing the idea of a non-symmetric distribution.
In the right panel, the situation is even more pronounced: extreme skewness (skewness = 3.2321) and very high kurtosis (15.5466) are observed, characteristics indicating a highly distorted distribution. The Anderson-Darling test p-value (<0.005) confirms the lack of normality. Again, the histograms and box plots reveal a strong concentration of values at the lower end and reat ispersion with multiple outliers. The confidence intervals for the mean (44,856,752 to 59,468,952) and for the median (34,530,278 to 44,261,350) are far apart, confirming an asymmetric distribution.
These results align with what was indicated by Ghasemi & Zahediasl (2012), who mentioned that statistical models applied to this type of data should consider robust techniques in the face of non-normality, such as logarithmic transformations or the use of non-parametric models, as classical assumptions of normality are not met. To further confirm the non-normality of the data, probability plots between clicks and impressions were also generated.

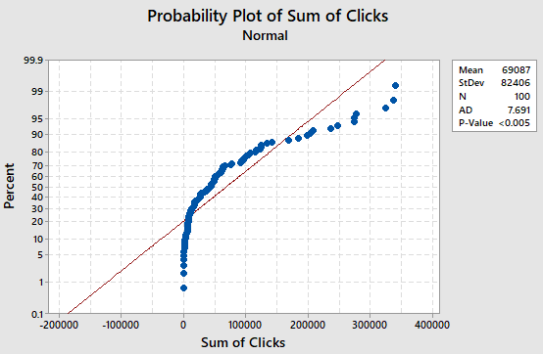

Illustration 8: Probability plot of "Sum of Clicks" in relation to a normal distribution. (Author's elaboration, 2025)

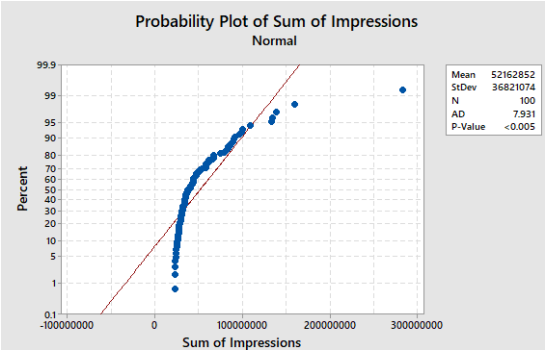

Illustration 9: Probability plot of "Sum of Impressions" in relation to a normal distribution. (Author's elaboration, 2025)

In Illustrations 8 and 9, an alignment of the points around the diagonal line would indicate an approximately normal distribution. However, the plots show notable deviations, especially in the tails, suggesting the presence of outliers and possible skewness in the data. This observation is reinforced by the normality test results, whose p-values (<0.005) indicate a very low probability that the analyzed variables are normally distributed. As noted by García Merino et al. (2023), verifying statistical assumptions, such as normality, is essential to ensure the validity of analyses applied in scientific research. In this case, the obtained results justify the use of non-parametric statistical methods or appropriate transformations that allow the assumptions to be met.

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	54074	16889	3.20	0.002	
Device Type					
Smartphone	17683	16758	1.06	0.294	1.01
Amount of Biomedical information					
High	-5246	25389	-0.21	0.837	1.19
Medium	12288	18188	0.68	0.501	1.20

Table 3: Regression model on device type and amount of information per creative

The regression analysis evaluated the impact of device type and the amount of biomedical information on the number of impressions. The constant coefficient, which represents the baseline level of impressions for creatives with low biomedical information viewed on non-smartphone devices, was statistically significant (Coef = 54,074, p = 0.002), indicating a reliable estimate for this reference group. In contrast, none of the other predictors were statistically significant. Creatives viewed on smartphones showed an average increase of 17,683 impressions compared to non-mobile devices, although this effect was not statistically significant (p = 0.294). Similarly, creatives with a high level of biomedical information showed a non-significant decrease of 5,246 impressions (p = 0.837), while those with a medium level showed a non-significant increase of 12,288 impressions (p = 0.501), both compared to the low information group. These results suggest that neither the type of device nor the amount of biomedical information has a significant or statistically supported impact on impressions within this dataset. These interpretations follow the common guidelines of linear regression analysis described by Montgomery et al. (2012).

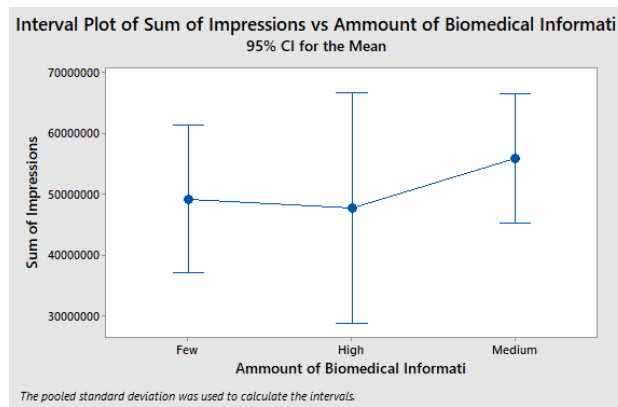

Illustration 10: ANOVA – Sum of impressions vs. Amount of biomedical information. (Author's elaboration, 2025)

In Illustration 10, the first ANOVA test allowed comparison of the total impressions across information levels (high, medium, and low). Although creatives with low information visually showed more impressions, the difference between groups was not statistically significant. Gravetter & Wallnau (2014) mention that this lack of statistical significance reinforces the idea that other factors beyond medical content may be influencing the visibility of ads.

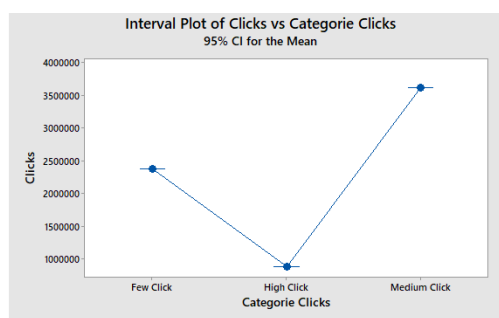

Illustration 11: ANOVA – Impressions vs. Amount of biomedical information. (Author's elaboration, 2025)

Regarding the interaction between biomedical information and the average number of impressions in Illustration 11, the observed pattern was consistent: creatives with less information generated more clicks on average. This trend suggests greater effectiveness of simpler messages or those with less technical jargon, possibly because they facilitate understanding for the average user, as described by Petty & Cacioppo (1986). However, as with impressions, no statistical test was applied to confirm the significance of these differences.

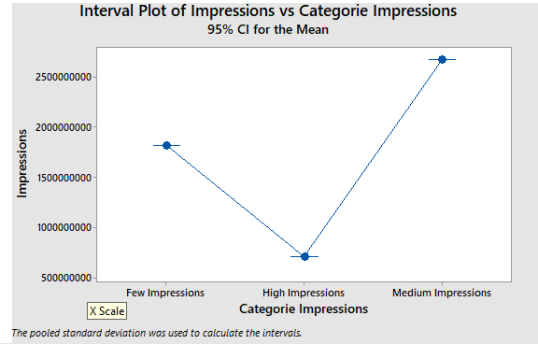

Illustration 12: ANOVA – Clicks vs. Amount of biomedical information. (Author's elaboration, 2025)

In Illustration 12, focused on clicks, non-significant differences were observed among the information levels. Although the data visually supports the hypothesis that less medical information improves user response, from a statistical standpoint the findings do not establish a causal relationship. The linear regression model on clicks, which considered both device type and amount of medical information, reinforced previous findings. Although a favorable trend was observed toward creatives with lower informational load, again, the coefficients were not significant (Tabachnick & Fidell, 2019).
In summary, the findings of this research did not demonstrate a statistically significant relationship between the amount of biomedical information in digital ads and user interaction, measured through impressions and clicks. Although creatives with lower information content showed a consistent trend of generating higher levels of interaction, these patterns did not reach statistical significance in the applied tests. This suggests that while there may be a preference for simpler and less technical formats, possibly due to their greater clarity and cognitive accessibility (Petty & Cacioppo, 1986), other contextual factors may be influencing user response, such as visual format, segmentation, or ad placement. Therefore, although the visual results partially support the proposed hypothesis, it is not possible to affirm a direct or conclusive relationship between biomedical informational density and interaction, which coincides with what was indicated by Gravetter & Wallnau (2014) regarding the importance of considering multiple variables in the analysis of digital consumer behavior.

4. Discussion

This research aimed to analyze user interaction with digital biomedical content using statistical techniques, with the goal of optimizing the dissemination of health information and improving its accessibility. From a quantitative perspective, and through tools such as regression, ANOVA, and descriptive analysis, patterns of interaction were explored in relation to variables such as the amount of biomedical information presented and the type of device used.

Overall, the findings suggest several relevant points. First, although ads with a lower amount of biomedical information generated more impressions and clicks on average, these differences were not statistically significant. This aligns with Petty and Cacioppo's (1986) Elaboration Likelihood Model (ELM), which posits that the central route of persuasion is activated mainly when consumers have high motivation and ability—conditions that are not always met in digital environments. Instead, users may rely on mental shortcuts (heuristics) to make quick decisions, favoring simpler content (Hoyer et al., 2019).

Second, regression analysis revealed no significant impact of device type on interaction rates, suggesting that content-related variables, rather than the delivery channel, play a more decisive role in ad effectiveness. This result is consistent with prior studies highlighting the importance of design, message clarity, and brand trust (Smith et al., 2021).

Third, normality tests showed that the data did not follow a normal distribution, which made it necessary to consider robust statistical approaches, as recommended by Ghasemi and Zahediasl (2012). This emphasizes the importance of verifying statistical assumptions in quantitative research, particularly when dealing with large datasets from digital environments.

Although the results were not conclusive in terms of statistical significance, they do reveal valuable patterns that can guide future decision-making in health communication campaigns. As Hernández and Mendoza (2018) note, discussion should interpret findings in light of prior knowledge and propose new courses of action.

From an alternative perspective, the preference for less technical ads could reflect users' cognitive load limitations when navigating digital spaces with competing information. Likewise, the lack of device-related effects might be due to the increasing homogenization of user experiences across platforms, making interface design and message framing more influential than the medium itself.

The study's strengths include the use of multiple statistical techniques, a focus on both content and delivery variables, and an empirical foundation for optimizing biomedical ad strategies. Limitations include the absence of qualitative measures to capture user perceptions, the reliance on a single dataset, and the fact that non-normality restricted the applicability of some statistical methods.

In terms of applications, the findings suggest that health communication strategies should prioritize clarity and simplicity in message design, especially for audiences without a strong technical background. For the discipline, this reinforces the need to combine communication theory with data-driven approaches in digital health promotion.

Finally, future research could expand on this work by integrating mixed methods to explore both behavioral patterns and user perceptions, employing A/B testing with varying technical levels, and incorporating qualitative metrics such as perceived credibility and message recall. These efforts would help build a more comprehensive understanding of how biomedical information presentation influences user engagement in the digital space.

4.1 Conclusions

- Biomedical ads with a lower amount of information generated, on average, a higher number of impressions and clicks, although these differences were not statistically significant.
- The type of device used by the user (smartphone, desktop, etc.) did not have a significant effect on interaction rates.
- The lack of normality in the data highlights the importance of applying robust statistical techniques or transformations when necessary.
- The research confirms that how biomedical information is presented—beyond its amount—can influence user response and should be carefully considered in future campaigns.

4.2 Recommendations

- For biomedical advertising campaign designers, prioritize visually attractive messages, concise wording, and clear language to capture the attention of users who may not have technical health knowledge.
- Consider the use of mixed quantitative and qualitative methods to explore both behavioral patterns and users' subjective perceptions.
- Implement A/B testing in digital marketing companies with varying levels of technicality in ads, evaluating not only clicks but also metrics such as dwell time or conversion rate.
- Integrate more qualitative and contextual metrics to complement traditional measures such as impressions or clicks, thus creating a more complete view of communicative impact and further optimizing digital ad spaces that significantly influence today's consumers.

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